# Hemodynamic Normals

Normal adult values, key formulas and how to read the waveforms behind them

**Medic Basics**
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## PRESSURE NORMALS

*Typical adult ranges.*

| PARAMETER | NORMAL | UNITS |
|---|---|---|
| **HR** | 60–100 | beats/min |
| **MAP** | 70–105 | mmHg |
| **CVP / RAP** | 2–6 | mmHg |
| **RV** | 15–25 / 0–8 | mmHg |
| **PA** | 15–25 / 8–15 | mmHg |
| **PA mean** | 10–20 | mmHg |
| **PAOP (wedge)** | 6–12 | mmHg |

Preload: CVP (right), PAOP (left). Afterload: PVR (right), SVR (left). Pulmonary hypertension: mean PA above 20 mmHg since WSPH 2018; older texts use 25 or more.

## FLOW, RESISTANCE AND OXYGEN

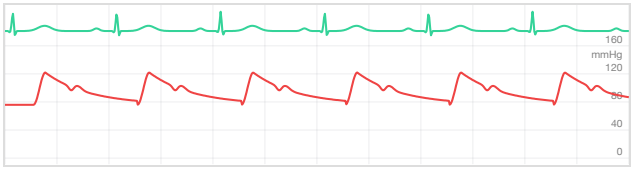
*1 WU = 80 dyn·s/cm$^5$; PH guidelines call PVR above 2 WU (160) high.*

| PARAMETER | NORMAL | UNITS |
|---|---|---|
| **CO** | 4–8 | L/min |
| **CI** | 2.5–4.0 | L/min/m² |
| **SV** | 60–100 | mL/beat |
| **SVI** | 33–47 | mL/m²/beat |
| **SVR** | 800–1200 | dyn·s/cm$^5$ |
| **SVRI** | 1970–2390 | dyn·s/cm$^5$·m² |
| **PVR** | 100–250 | dyn·s/cm$^5$ |
| **SvO2** | 60–80 | % |
| **ScvO2** | About 70 | % |
| **DO2** | 950–1150 | mL/min |

- MAP = (SBP + 2 × DBP) ÷ 3
- CO = HR × SV ÷ 1000 (L/min); CI = CO ÷ BSA
- SVR = (MAP − CVP) × 80 ÷ CO
- PVR = (mean PA − PAOP) × 80 ÷ CO
- CaO2 = 1.34 × Hb × SaO2 (decimal) + 0.003 × PaO2 (17–20 mL/dL)
- DO2 = CO × CaO2 × 10
- Coronary perfusion = DBP − PAOP (60–80 mmHg)

**EXAM TIP** Low SvO2 or ScvO2: delivery is not meeting demand; check CO, hemoglobin, SaO2 and oxygen consumption. ScvO2 runs slightly below SvO2 in health, about 7% above it in unstable patients.

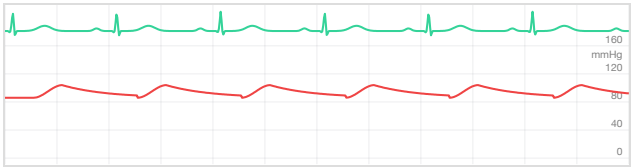
## ARTERIAL LINE AND SQUARE-WAVE TEST




Normal: brisk upstroke, systolic peak, dicrotic notch (aortic valve closure).

Square-wave test: snap the fast flush and count oscillations before the tracing returns.

| RESULT | OSCILLATIONS | ERROR |
|---|---|---|
| **Optimal** | 1.5–2 | None |
| **Overdamped** | Fewer than 1.5 | False low SBP, high DBP |
| **Underdamped** | More than 2 | False high SBP, low DBP |



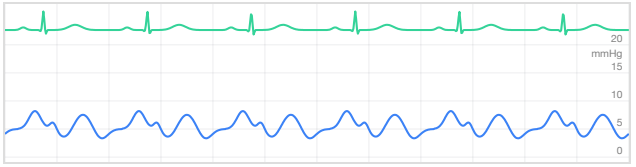

Overdamped: slurred upstroke, no notch, narrow pulse pressure. Look for air, clots, kinks or a soft flush bag (keep 300 mmHg).




Underdamped: overshoot spike with ringing; systolic reads falsely high. Check for long or added tubing.

**EXAM TIP** MAP stays the most reliable number on a damped line; confirm with a cuff while you fix it.

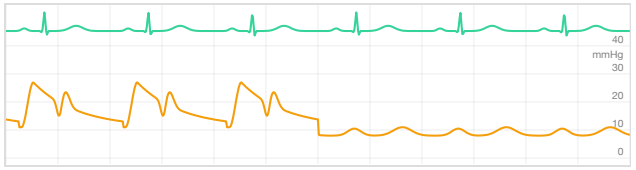
## CVP AND PA CATHETER WAVEFORMS




CVP: a wave after P (atrial kick), c after QRS (tricuspid closure), v just after T (atrial filling).




Insertion: RA, RV (diastolic near 0), PA (diastolic step-up, dicrotic notch), then wedge.




Balloon up: PA trace drops to wedge, mean below PA diastolic.

- PAOP estimates LA pressure and LVEDP; it reads high with mitral disease or high PEEP, low with a stiff LV.
- PAOP sits a few mmHg below PAD; a wide gap suggests high PVR. Small gap, stable patient: trend PAD and wedge less.
- Cannon a waves: AV dissociation. No a waves: atrial fibrillation.
- Giant v waves: tricuspid (CVP) or mitral (wedge) regurgitation.

**SAFETY** Check the PA trace before inflating; if damped, don't. Inflate slowly and stop at wedge, never past the 1.5 mL marked; wedge at under 1.25 mL suggests distal migration. Wedge 2 breaths or less (10–15 s), deflate passively, never flush while wedged. Wedge or RV trace with balloon down: catheter has moved (wedge risks infarction). Act per protocol.

## READING THE NUMBERS

- Level the air-fluid interface to the phlebostatic axis (4th intercostal space, midway front to back); re-level when position changes.
- Zero to air. Head of bed 0–60° is fine once leveled.
- Read at end-expiration on a frozen or printed trace, not the digital number: just before the inspiratory dip (spontaneous) or rise (ventilator).
- Read CVP and PAOP at the mean of the a wave.

**SAFETY** A transducer below the phlebostatic axis reads falsely high; above it reads falsely low, about 2 mmHg per inch.

## FLUID RESPONSIVENESS

- Responder: SV or CO rises more than 10% with a fluid challenge or passive leg raise (PLR).
- PLR: from 45° semi-recumbent, tilt the bed to trunk flat, legs up. Judge by SV or CO within 1–2 minutes, not BP.
- PPV above 13% or SVV above about 12% predicts response.
- PPV and SVV need controlled ventilation, no spontaneous breaths, regular rhythm, tidal volume about 8 mL/kg IBW, closed chest, normal abdominal pressure.
- SSC 2026: dynamic measures over physical exam or static values alone. Septic shock: initial MAP 65 mmHg (60–65 if 65 or older).

**EXAM TIP** CVP and PAOP are static and poorly predict fluid response; exams favor PLR, SVV and PPV. For shock profiles, see Shock Types. Study values only; follow your institution's protocols.

**Sources** (20) are listed with links at medicbasics.com/sheets/hemodynamic-normals.html. For education and exam preparation only. Not medical advice; follow current guidelines and your institution's protocols. Medic Basics is an independent study aid not affiliated with any certifying body.